

Dear editors,

Re: "Unsupervised Functional Stage Detection From EEG Using Topological Data Analysis"

Manuscript reference: MBEC-D-25-03498R2

We appreciate the further comments and suggestions from the reviewers, and the time devoted to this round of evaluation. We have addressed all remaining minor issues in the revised manuscript. Below, we provide a point-by-point response, outlining the corresponding revisions.

[Reviewer #4] In the previous review I pointed to two studies demonstrating the value of shorter epochs in insomnia and sleep-fragmentation research and one examining how different PSD estimators affect EEG analysis. These suggestions appear to have helped shape the revisions, yet the papers themselves are neither referenced nor discussed in the manuscript. Given that they contributed to improving the work, I wonder why they were omitted from the reference list and from the relevant passages in the Methods and Discussion. I would be grateful if the authors could reconsider this point.

We apologize for the omission. We have added the suggested studies to the reference list and cited them in the passage motivating the 15-second epochs in Section 3.2 of the revised manuscript.

[Reviewer #4] Feature selection and final evaluation still rest on the same objective. A brief nested or leave-one-subject-out check, or at least a clearer statement of the residual risk, would further strengthen the claim of robustness.

We have expanded the corresponding paragraph in the Discussion to state the residual risk more explicitly. The revised text clearly acknowledges the need for a more formal evaluation via a nested protocol or a leave-one-out check and flags this as a concrete direction for future work.

[Reviewer #4] Selecting the best-fitting subset of ten expert stages can inflate FMI. A short sensitivity note or a more explicit caution in the Discussion would improve transparency.

We added a more explicit caution in Section 3.2, immediately after the description of the matching procedure. We explicitly acknowledge that the FMI should be read as an upper bound on structural agreement rather than as a conventional sleep-staging score.

[Reviewer #4] ICA selection criteria and the referencing scheme remain largely deferred to the earlier SDA paper. A concise summary in Section 2.1 would make the present manuscript more self-contained.

We have expanded the preprocessing description in Section 2.1 to make the manuscript more self-contained. The revised passage now briefly outlines the referencing scheme and the ICA selection criteria. The reference to the original SDA paper is retained for the exhaustive parameter list.

[Reviewer #5] I have one suggestion. The authors only tested their framework on one dataset. I think they should say that this is a limitation of their study. It would be good if they talked about how they need to test it on datasets or with different groups of people, in the future. This would make their results more believable.

[Reviewer #8] Recommend adding a short limitations paragraph about generalizability (since only healthy aging subjects were used).

We have added an explicit note on this limitation in the final paragraph of the Discussion. We now explicitly frame the observed improvements within the two datasets rather than as general judgments, and we call for future validation on other clinically interesting data.

[Reviewer #8] Correct the duplicated "is is" in the abstract.

Corrected. The abstract now reads "EEG segmentation is still largely performed manually by specialists."

[Reviewer #8] Encourage clearer visual presentation (a figure comparing epoch length performance) to complement tables.

We considered adding such a figure, but on reflection it merely restated the numbers already in Table 2 without adding new information. A genuinely complementary visualization would be to show the distributions behind the statistics, which is already provided in Supplementary Materials E, referenced in Section 3.2. We believe the manuscript already covers the epoch-length comparison, and additional figures in the main text would be redundant.

Once again, we are grateful for the valuable suggestions and thank you and the reviewers for the time dedicated to our manuscript.

Sincerely yours,

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